

Project 1, Session 5: Data visualization II: quantitative representations

- Paired team review
- Appropriate choice of scale
- Displaying and interpreting distributions
- Preparation for presentations

Team discussion

Paired teams: 1-4-5, 2-6, 3-9, 7-8

- How did you assign each PTEN variant to a region?
- How did you summarize the number of variants, included sites, and total number of sites in each region? How do these compare across regions?
- How did you compute median scores -- across variants and across median scores for each included site -- for each region, and how do you interpret them?
- How did you visualize and interpret score distributions across regions?
- What variables did you change to test the sensitivity of the results, and how did these changes affect the conclusions?

When presenting, make sure to **clearly describe your code** and what it does.

Figures as quantitative representations of data

We want to present information faithfully, with context.

Often used to support a conclusion or interpretation, but the chain of logic needs to be
data → interpretation → representation rather than interpretation → data → representation.

Quantitative presentation is mostly orthogonal to aesthetics, but for both, good design makes data easier to understand and interpret.

Axis limits change visual contrast

```
show_plot(s4_axis_limits)
```

Which scale is most relevant? Context is important.

Linear differences and log ratios

What is the natural scale of the data?

```
show_plot(s4_log_scale)
```

Distinguishing missing data from zeros

```
show_plot(s4_missingness)
```

Regional medians and observations

Showing only summary statistics can hide important variation.

```
show_plot(s5_summary_distribution)
```

More detailed representations can show spread in scores, range of variation, and different sample sizes.

Histograms

Histograms are commonly used to visualize distributions by counting the number of values that appear within a given interval of values.

While apparently simple, histograms can be (notoriously) sensitive to display choices.

Bin width and apparent structure

How many peaks would you describe in each panel?

```
show_plot(s5_histogram_bins, mode="width")
```

The starting edge matters too

The bin widths now match, but the interval range has changed. This shifts the apparent location of the peak and its height.

```
show_plot(s5_histogram_bins, mode="origin")
```

Counts and proportions

Do you want to compare the number of positions, or the shape within each region?

```
show_plot(s5_region_histograms)
```

When displaying multiple histograms together, it's useful to use the same bins and describe sample sizes.

Alternative: a distribution without bins

Cumulative distributions can also be used to visualize data. No bin choices, but a bit less intuitive?

```
show_plot(s5_ecdf)
```

Limited range can affect interpretations of the data

```
show_plot(s5_full_range)
```

Quantitative representations with color

When using color to convey information across related plots, use a single scale and reference for consistency.

```
show_plot(s5_shared_color_limits)
```

Example: filtering PTEN data to require $\text{expts} \geq 6$

Comparison across sites shows that most sites have consistent results with different filtering criteria, but not all.

```
show_plot(s5_filter_sensitivity)
```

242 positions remain eligible; 75 lose eligibility.

Project 1 presentations

Your team will define and investigate one question that extends the PTEN analyses that we've performed together.

Next class, each team will present their results, including

- A definition of the question
- How you approached it, including **at least one** explicit code example with an accompanying explanation
- A brief summary of your results and conclusions
- Plan for **5-7 minutes** + brief questions
- Presentations can use slides and the whiteboard (encouraged for describing the problem and approach!)

Team activity: choose a new way to extend your analysis

Prediction sheet:

- Define the question or goal
- How will you approach it? Does this require new data or new analysis?
- What are your anticipated results?
- How will you ensure that your analysis is correct?